Coral Reefs (2015) 34:81–86
DOI 10.1007/s00338-014-1224-4

NOTE

Clustered parrotfish feeding scars trigger partial coral mortality of massive *Porites* colonies on the inshore Great Barrier Reef

J. Q. Welsh · R. M. Bonaldo · D. R. Bellwood

Received: 2 April 2014 / Accepted: 30 September 2014 / Published online: 12 October 2014
© Springer-Verlag Berlin Heidelberg 2014

Abstract Coral predation by parrotfishes can cause damage to coral colonies, but research into the dynamics of their feeding scars on Indo-Pacific corals is limited. We monitored feeding scars of the parrotfish *Chlorurus microrhinos* on massive *Porites* colonies at Orpheus Island (inshore Great Barrier Reef) over 4 months. Of the 30 marks monitored, 11 were single feeding scars, which all healed completely. The remaining 19 feeding marks consisted of clusters of scars. Eight began to recover, while 11 increased in size by 1,576 ± 252 % (mean ± SE). A logistic regression predicted that a single feeding scar on a *Porites* colony had a 97 % probability of healing; however, where more than three feeding scars were present, this dropped below 50 %. As excavating parrotfishes in the Indo-Pacific often take multiple focused bites, they may have a significant impact on the growth and mortality of massive *Porites* colonies at Orpheus Island.

Keywords Coral reefs · Corallivory · Coral predation · *Chlorurus microrhinos* · *Porites* · Grazing scars

Communicated by Handling Editor Simon Davy

J. Q. Welsh (✉) · D. R. Bellwood
School of Marine and Tropical Biology, Australian Research Council Centre of Excellence for Coral Reef Studies, James Cook University, Townsville, QLD 4811, Australia
e-mail: justin.welsh@my.jcu.edu.au

R. M. Bonaldo
Departamento de Ecologia, Instituto de Biociências, Universidade de São Paulo, São Paulo, SP, Brazil

Introduction

In recent years, corallivory by fishes has been identified as a source of significant coral mortality (e.g. Bruckner and Bruckner 1998; Rotjan and Lewis 2008; Cole et al. 2011). These corallivorous fishes primarily target fast growing coral species, especially the Acroporidae and Pocilloporidae (Cole et al. 2008; Bellwood et al. 2010), which have the potential to rapidly regenerate following damage from predation (Cole et al. 2011). In contrast, relatively few Indo-Pacific corallivores prey on slow growing massive corals, such as *Porites*. However, some fishes do target *Porites,* and this predation may significantly affect coral population dynamics (Rotjan and Lewis 2008; Jayewardene et al. 2009; Bonaldo and Bellwood 2011).

Of the fish species that feed on corals, the Chaetodontidae, Labridae (including parrotfishes) (Bonaldo and Bellwood 2011) and Tetraodontidae (Jayewardene et al. 2009) play the largest role, with individual fish removing significant quantities of coral tissue when feeding. In recent years, parrotfish corallivory has been increasingly recognized as an important factor influencing coral population demographics. Although parrotfishes have long been regarded as important coral predators in the Caribbean (e.g. Bruckner and Bruckner 1998; Mumby 2009), they have only recently been found to play a similar role on the Great Barrier Reef (GBR) (Bonaldo and Bellwood 2011; Bonaldo et al. 2011, 2012, 2014; Cole et al. 2011).

The effect of parrotfish grazing on coral colonies can be highly variable and dependent on parrotfish feeding mode (reviewed by Bonaldo et al. 2014). For example, scraping species only remove the outermost layer of tissue, doing little damage to the corallite (Cole et al. 2008; Bonaldo and Bellwood 2009; Bonaldo et al. 2011, 2012). In contrast, excavating parrotfish species have more robust jaws than

their scraping counterparts (Bellwood and Choat 1990) and are able to penetrate deeper into the coral, removing entire polyps as well as their corallites (Bruckner and Bruckner 1998; Cole et al. 2008; Bonaldo et al. 2011).

A recent study on the short-term fate of parrotfish feeding scars on massive *Porites* colonies from a reef in the mid-shelf GBR found that the relatively frequent scraping scars were usually recovered, whereas excavating scars remained unchanged for at least 2 months (Bonaldo et al. 2011). In the Caribbean, long-term studies of parrotfish corallivory observed low rates of recovery, especially where the underlying coral skeleton is damaged (Frydl 1979; Rotjan and Lewis 2005). It was therefore hypothesized that coral predation by excavating parrotfishes, especially in cases of coral skeleton damage, may cause long-term damage to some coral colonies (Rotjan and Lewis 2008; Bonaldo et al. 2011), potentially altering the structure and morphology of corals such as *Porites*. As these corals are key ecosystem engineers on coral reefs, changing their growth patterns may influence the physical structure of the reef (Bonaldo et al. 2011; Goatley and Bellwood 2011).

However, the nature of parrotfish bites can range from a single bite taken from a colony to a more dramatic feeding event, in which several bite marks are taken from a single area (Bruckner et al. 2000; Bonaldo et al. 2011). This 'focused biting' behaviour (sensu Bruckner et al. 2000) is usually displayed by excavating species and, in extreme cases, has been shown to cause total colony mortality as parrotfish consume nearly all living tissue from the colony (Bruckner and Bruckner 1998). Even if not all the living tissue is removed from a coral colony, studies from the Caribbean have documented whole-colony mortality following intense and focused feeding by corallivorous parrotfishes (Bak and Steward-Van Es 1980; Rotjan and Lewis 2005). Despite the potential severity of this behaviour, studies on the impact and rate of recovery of sub-lethal parrotfish feeding scars from 'focused biting' behaviour have largely focused on Caribbean reefs and data remain rare for Indo-Pacific reefs. In this study, therefore, our aim was to monitor excavating parrotfish feeding scars on *Porites* colonies from a reef within the inshore GBR to assess the potential for recovery, and to verify if 'focused biting' by an excavating parrotfish species, *Chlorurus microrhinos*, can cause localized coral mortality at this reef. We also use a logistic model to predict the recovery potential of feeding scars based on the number of bites taken from a single area.

Materials and methods

This study was conducted from July to October on the fringing reefs of Pioneer Bay, on the leeward side of Orpheus Island (18°350′S, 146°200′E), an inshore granitic island on the GBR. Two sites were selected in which to locate and monitor feeding scars. Fresh feeding scars were located by following individual *C. microrhinos* of approximately 40–45 cm (total length) until they bit a massive *Porites* colony. Once the individual had finished feeding, the grazing scar (Fig. 1a) or scar cluster (Fig. 1b) (collectively referred to as feeding marks) was tagged using tape tied to a nearby outcrop. Only one feeding damage event was considered per coral colony (with detailed notes of the location of each mark to ensure the same mark was tracked over time). Photographs were taken of each feeding mark from a fixed distance with a scale for size reference. Scars were only considered for the study when present on a relatively planar surface of the coral colony. Photographs were also taken of the entire surface of each predated coral colony. These photos were combined to estimate the 2D area of each of the coral colonies. The predated colonies were broadly similar in size and ranged from 440.9 to 8,386.7 cm^2 (average of 1,981 ± 334.4 cm^2; ±SE). As fish sizes, coral species and colony locations were all similar; the individual scars were also of a similar size and depth, measuring 5.81 ± 1.91 cm^2 (±SE; range of 4.18–6.19 cm^2) and 0.13 ± 0.01 cm deep (range of 0.02–0.18 cm). A photograph was taken of each feeding mark immediately after feeding, and in each subsequent month (a total of four observations). All monitored coral colonies were located on the reef crest approximately 0.5–1 m deep (below chart datum). Observations were restricted to scars on massive *Porites* as this form is abundant at the study site, which exhibits clearly distinguishable feeding scars, and is often bitten by *C. microrhinos* (Bonaldo et al. 2012; Welsh and Bellwood 2012). Because of the difficulty in distinguishing massive *Porites* species on the GBR, colonies were classified as 'massive *Porites* spp.'.

Data analysis

Once photographed, the perimeter and surface area of the feeding marks, the associated coral colony area, and the number of scars in each bite cluster were calculated using the software ImageJ64 (Abramoff et al. 2004). At the end of the study period, the fate of each feeding mark was classified as either having recovered (in which case polyps were evident across the scar surface, and the colour of the coral tissue was indistinguishable from the rest of the colony), decreased or expanded (scars still with exposed corallites and/or algal growth). No feeding marks had remained the same at the end of the study. To compare the frequency of scar cluster recovery between sites, a chi-squared goodness of fit test was applied. Also, two binary GLM were used to model the relationship between the response variable of 'fate of feeding damage' and either (1) the initial number of feeding scars in a bite cluster and

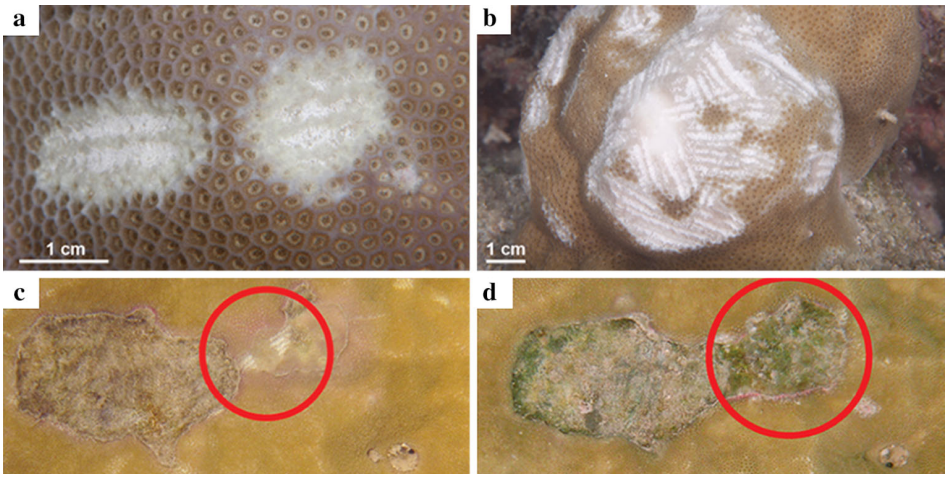

Fig. 1 **a** A single parrotfish (*C. microrhinos*) feeding scar, **b** a scar cluster on the surface of a *Porites* colony and **c** a single scar resulting in **d** partial coral mortality surrounding a scar

colony size or (2) the initial area to perimeter ratio of the feeding mark. For these analyses, binary data were required. Therefore, feeding marks were classified as either recovering (healed completely or decreased in size) or expanded (increased). For further investigation, two separate logistic regressions (Nagelkerke R^2) were also used to quantify the recovery potential: one of them with number of feeding scars and the other with the area to perimeter ratio of the initial feeding mark. Tests were run in R 3.0.1 for Mac OS×. Nagelkerke R^2 was calculated with 'rms' package (Harrell Jr 2014), and data plots were built with ggplot2 extension for R (Team RDC 2008).

Results and discussion

This study presents empirical evidence of the fate of excavating parrotfish feeding scars on *Porites* colonies from a reef on the inshore GBR. We found that feeding marks on *Porites* colonies caused by parrotfish focused biting is long-lasting and, in some cases, expanded over time. Of the *C. microrhinos'* feeding marks on massive *Porites*, 11 were composed of a single feeding scar and 19 were scar clusters from focused biting ($n = 15$ per site); sites were pooled for analysis as no significant difference was found for the recovery potential between sites ($\chi^2_1 = 1.20$, $p = 0.27$). Damage from a single feeding scar showed signs of recovery by the second month of monitoring, and, in all cases, the damage had completely regenerated by the third month. From the grouped scars, six recovered completely, two decreased in area (by 44 and 61 %) and 11 increased in size. Mean increase in the mark area was 1,576 ± 252 % (±SE). This expansion in the damaged area was marked by coral bleaching at the margins, then tissue loss along the margin and algal invasion. These results suggest that the number of bites taken by *C. microrhinos* on massive *Porites* directly influences the probability and extent of local colony mortality (Fig. 2).

A number of studies have shown that excavating parrotfish species commonly feed by 'focused biting' (e.g. Bruckner and Bruckner 1998; Bonaldo et al. 2011). This behaviour, in conjunction with the fact that *C. microrhinos* focuses its feeding on small areas within its territories at the study site (Bonaldo et al. 2012, Welsh and Bellwood 2012), may result in scars concentrated in areas of the reef with intense parrotfish feeding (e.g. Bonaldo et al. 2012), causing long-lasting damage to *Porites* colonies. This is evidenced by the binary GLM, as well as the logistic regression, which found a significant effect of number of scars and surface to perimeter ratio in the initial predation event, and the subsequent fate of the mark on the colony (Table 1).

Colony size and chronic damage from repeated feeding events of corallivores have been suggested to significantly influence the capacity of a coral colony to recover from damage (Rotjan and Lewis 2005). However, the influence of colony size on recovery potential appears to be most marked in very small colonies (by GBR standards), as no effect of colony size was detected in the present study (Tables 1, 2). As feeding damage represented, on average, less than 23.1 % (±5.8) of the corals planar area, it is unclear whether coral size may influence the ability of corals to recover at this site. In the Caribbean, colony size was considered a significant factor predicting colony recovery (Frydl 1979) although this may only be significant in colonies <80 cm^2 (Bak and Steward-Van Es 1980), which were not included in the present study.

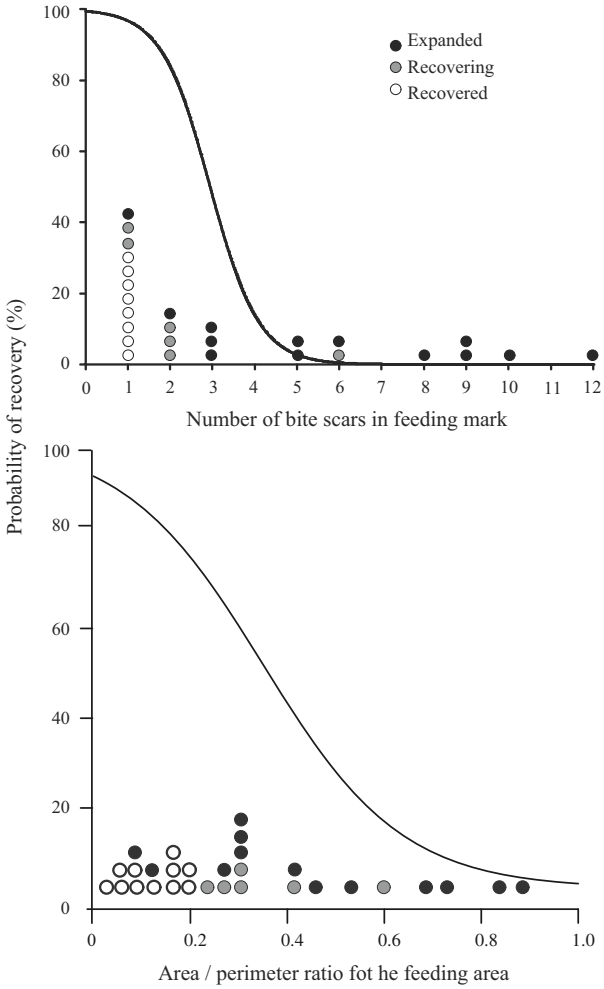

Fig. 2 Logistic model of the relationship between the probability of feeding scar recovery, the number of feeding scars initially recorded ($y = e^{5.11 - 1.74(x)}/(1 + e^{5.11 - 1.74(x)})$ (*top*) and the initial area to perimeter ratio ($y = e^{6.94 - 2.42(x)}/(1 + e^{6.94 - 2.42(x)})$ (*bottom*). Each dot marks the final status of a scar after 4 months of monitoring, being either fully recovered (*open*), reduced (*grey*) or expanded (*black*) ($n = 30$)

Table 1 Results of the binary General Linear Model assessing the influence of initial scar count and colony size on the fate of feeding scars ($n = 30$)

	Estimated std.	Error	z value	p
Intercept	6.49	2.87	2.26	**0.024**
Number of scars	−2.76	1.42	−1.94	**0.043**
Colony size	0.0005	0.0009	0.61	0.54

Dispersion parameter was 1 with AIC value of 18.68. Significant values ($p < 0.05$) presented in bold

In the present study, when only the number of scars was considered, the logistic regression was a significant predictor of the capacity for colony recovery ($\chi^2 = 26.206$, $df = 1$, $p < 0.05$) and predicted 79 % of the variability in the relationship between the fate of a feeding mark and the number of bite scars in the feeding mark (Nagelkerke $R^2 = 0.79$). The AIC values for this model were only 1.6 units lower than the model that also considered colony size as a potential influence on recovery potential. Therefore, colony size was not a significant predictor ($p = 0.70$). Based on the model, a single feeding scar on the *Porites* colonies has a 97 % probability of healing; however, the probability of recovery dropped sharply, to below 50 %, if more than three feeding scars were present in a single mark (Fig. 2). For the area to perimeter ratio of initial feeding marks, the logistic regression was also significant ($\chi^2 = 11.07$, $df = 1$ and $p = 0.0009$), but predicted less of the variability between the fate of a feeding mark and the number of bite scars in the feeding mark (Nagelkerke $R^2 = 0.41$). This model predicted that bite marks with small area to perimeter ratio have a high chance of healing, but the probability of recovery drops below 50 % if the area of the mark is 40 % larger than the perimeter (area vs. linear values notwithstanding) (Fig. 2).

Table 2 Results of the binary General Linear Model assessing the influence of initial area to perimeter ratio and colony size on the fate of feeding scars ($n = 30$)

	Estimated Std.	Error	z value	p
Intercept	2.52	0.96	2.62	**0.009**
Area to perimeter ratio	6.92	2.77	2.54	**0.01**
Colony size	7.77	2.44	0.32	0.75

Dispersion parameter was 1 with AIC value of 35.89. Significant values ($p < 0.05$) presented in bold

It has been suggested that, by targeting the growth tips of acroporid corals, tetraodontids and the parrotfish *Bolbometopon muricatum* significantly modify coral morphology and the reef structure (Sano et al. 1984; Bellwood et al. 2003; Cole et al. 2008). More recently, studies on *Porites* predation on Lizard Island, a mid-shelf GBR reef, have also suggested that excavating parrotfishes can change the distribution and shape of *Porites* colonies (Bonaldo and Bellwood 2011; Bonaldo et al. 2011). The present study reinforces these suggestions. By causing large areas of damage to *Porites* colonies, feeding scars have the potential to expand resulting in partial colony mortality or even fragmenting the colony into two distinct genets. As such, *C. microrhinos* may have a significant effect on these massive, reef-building corals.

Vulnerability of corals to focused feeding coupled with the behavioural inclination of *C. microrhinos* to feed on small areas within its territory (Welsh and Bellwood 2012) may create a damaging positive feedback for coral colonies. Studies of a number of coral reefs around the world have shown an inverse relationship between coral abundance and bite frequency for some coral species (e.g.

Jayewardene et al. 2009; Bonaldo and Bellwood 2011; Burkepile 2012). On degraded reefs with fewer coral colonies, it is possible that the remaining colonies will experience progressively heavier parrotfish predation pressure, thus decreasing coral health, fecundity and the overall resilience of coral populations (Szmant and Gassman 1990; Meesters et al. 1994; Shantz et al. 2011).

Because *C. microrhinos'* bite scars were monitored for just 4 months, the longer-term implications must be interpreted with care. For instance, a previous study in Okinawa reports that 29 out of 30 artificially made lesions on *Porites* completely recovered within 19 months, even though three scars expanded during the first 4 months (see van Woesik 1998 for further details). However, while the depth of the scars in van Woesik (1998) is not specified, they do appear to be shallower than ours. *C. microrhinos* bite scars at Orpheus Island are 4.86 ± 0.29 mm deep (mean ± SE) (Bonaldo and Bellwood 2009) and may remove the entire living tissue and polyps when inflicted (Bonaldo et al. 2011, 2014). This observation, in addition to the fact that 11 out of 30 scars expanded in our study, compared with just three out of 30 in van Woesik (1998), over the same time period, suggests that, even after longer periods of time, *C. microrhinos* bite marks may not always recover.

A number of studies on reefs around the world have reported limited recovery of coral colonies from scars inflicted by fish bites, particularly when this involves excavating or scraping fishes (e.g. Rotjan and Lewis 2005; Bonaldo et al. 2011, 2014; Denis et al. 2011). Although the recovery of single bites and small lesions (with low area to perimeter ratio) is commonly reported (e.g. van Woesik 1998; Sánchez et al. 2004; Bonaldo et al. 2011), larger deeper bites usually involve tissue loss with accompanying skeletal damage and are associated with lower regenerative success (Bak and Steward-Van Es 1980; Bak 1983; Croquer et al. 2002; Bonaldo et al. 2011). Partial mortality can impair growth and survival of remaining tissue, as regeneration requires the reallocation of resources for regeneration at the expense of colony growth (reviewed by Henry and Hart 2005); a process that is particularly energy-demanding for neighbouring polyps (Meesters et al. 1994; Henry and Hart 2005). Regeneration costs are highest in the area bordering live tissue, but regenerative capacity does not appear to change with colony size (Meesters et al. 1994; Oren et al. 1997). Therefore, although bigger colonies presumably have larger energy stores and more resources, the likelihood of recovery from damage is approximately equal among differently sized colonies.

Corallivory can also negatively affect coral reproductive potential, as energetically expensive reproduction is often compromised in favour of tissue regeneration processes (Szmant-Froelich 1985; Harrison and Wallace 1990; Van Veghel and Bak 1994). For this reason, considering the depth of bites inflicted by excavating parrotfish species, regeneration of the scars in the present study may represent a significant energetic cost to the coral.

Despite the negative effects of parrotfish corallivory, records of total colony mortality caused by parrotfishes are relatively rare (e.g. Bruckner and Bruckner 1998; Rotjan and Lewis 2005). Negative effects of parrotfishes on corals are largely restricted to particular coral species, within specific habitats (e.g. Littler et al. 1989; Rotjan and Lewis 2005; Mumby 2009; Bonaldo and Bellwood 2011). Although this predation may result in local declines, it may also support coral species richness within reef zones by decreasing the abundance of dominant coral species (e.g. Littler et al. 1989; Rotjan and Lewis 2005; Mumby 2009; Bonaldo and Bellwood 2011). The effects of parrotfishes on corals are complex. They may enhance coral settlement, but may increase juvenile mortality (Trapon et al. 2013) and change the growth rates, growth forms and survival of coral species.

This coral scarring by parrotfishes needs to be considered in the broader context of the functional roles of parrotfishes on coral reefs, which includes herbivory, bioerosion, sediment transfer, etc. (reviewed in Bonaldo et al. 2014). Through their important role in mediating coral–algal interactions in favour of corals (Hughes et al. 2007), parrotfishes, such as the excavating *C. microrhinos*, may ameliorate any negative impacts they may have on coral colonies (e.g. Littler et al. 1989; Bruckner and Bruckner 1998; Rotjan and Lewis 2005; Bonaldo et al. 2011). On balance, partial mortality and occasional mortality may be a small price to pay for the beneficial effects of parrotfishes in terms of herbivory and bioerosion.

On degraded reefs, however, this balance may change. It is possible that reduced coral cover may increase the per-colony rate of scarring by parrotfishes. Nevertheless, degraded reefs are usually characterized by reductions in parrotfish populations, especially of large excavating species (Bellwood et al. 2012). It is therefore likely that these reefs may exhibit reduced coral scarring while maintaining herbivory (a similar pattern seen on inshore coral reefs; Bonaldo et al. 2014). Cole et al. (2008) suggested that corallivory must be balanced by coral growth rates to maintain coral populations. However, when this balance is altered by environmental degradation, which slows coral growth (De'ath et al. 2009), it is possible that the negative impacts of parrotfishes may be exacerbated as coral growth slows, and corals become increasingly incapable of coping with ongoing fish predation.

Whether parrotfishes play a positive role in removing algae, enhancing coral recruitment and removing competitively dominant corals or a negative role by producing localized scars which may result in increased energetic costs, reduced reproduction or partial mortality (as demonstrated herein), it appears that the limited number of

parrotfish species which feed on massive *Porites* may play an important role in coral demographics, at least on some inshore reefs of the GBR.

Acknowledgments We thank the staff of Orpheus Island Research Station for invaluable field support; Y. Bosiger, D. Duchene, C. Goatley and M. Kramer for field assistance, and three anonymous reviewers for valuable suggestions. Financial support was provided by the Australian Research Council.

References

Abramoff MD, Magalhaes PJ, Ram SJ (2004) Image processing with ImageJ". Biophotonics International 11:36–42

Bak RPM (1983) Neoplasia, regeneration and growth in the reef-building coral *Acropora palmata*. Mar Biol 77:221–227

Bak RPM, Steward-Van Es Y (1980) Regeneration of superficial damage in the scleractinian corals *Agaricia agaricites* f. *purpurea* and *Porites astreoides*. Bull Mar Sci 30:883–887

Bellwood DR, Choat JH (1990) A functional analysis of grazing in parrotfishes (family Scaridae): the ecological implications. Environ Biol Fishes 28:189–214

Bellwood DR, Hoey AS, Choat JH (2003) Limited functional redundancy in high diversity systems: resilience and ecosystem function on coral. Ecol Lett 6:281–285

Bellwood DR, Hoey AS, Hughes TP (2012) Human activity selectively impacts the ecosystem roles of parrotfishes on coral reefs. Proc R Soc B Biol Sci 279:1621–1629

Bellwood DR, Klanten S, Cowman PF, Pratchett MS, Konow N, Van Herwerden L (2010) Evolutionary history of the butterflyfishes (f: Chaetodontidae) and the rise of coral feeding fishes. J Evol Biol 23:335–349

Bonaldo RM, Bellwood DR (2009) Dynamics of parrotfish feeding scars. Mar Biol 156:771–777

Bonaldo RM, Bellwood DR (2011) Parrotfish predation on massive *Porites* on the Great Barrier Reef. Coral Reefs 30:259–269

Bonaldo RM, Krajewski JP, Bellwood DR (2011) Relative impact of parrotfish feeding scars on massive *Porites* corals at Lizard Island, Great Barrier Reef. Mar Ecol Prog Ser 423:223–233

Bonaldo RM, Welsh JQ, Bellwood DR (2012) Spatial and temporal variation in coral predation by parrotfishes on the GBR: evidence from an inshore reef. Coral Reefs 31:263–272

Bonaldo RM, Hoey AS, Bellwood DR (2014) The ecosystem roles of parrotfishes on tropical reefs. Oceangr Mar Biol 52:81–132

Bruckner AW, Bruckner RJ (1998) Destruction of coral by *Sparisoma viride*. Coral Reefs 17:350

Bruckner AW, Bruckner RJ, Sollins P (2000) Parrotfish predation on live coral: "spot biting" and "focused biting". Coral Reefs 19:50

Burkepile DE (2012) Context-dependent corallivory by parrotfishes in a Caribbean reef ecosystem. Coral Reefs 31:111–120

Cole AJ, Pratchett MS, Jones GP (2008) Diversity and functional importance of coral-feeding fishes on tropical coral reefs. Fish Fish 9:286–307

Cole AJ, Lawton R, Pratchett M, Wilson S (2011) Chronic coral consumption by butterflyfishes. Coral Reefs 30:85–93

Croquer A, Villamizar E, Noriega N (2002) Environmental factors affecting tissue regeneration of the reef-building coral *Montastrea annularis* (Faviidae) at Los Roques National Park, Venezuela. Rev Biol Trop 50:1055–1065

De'ath G, Lough JM, Fabricius KE (2009) Declining coral calcification on the Great Barrier Reef. Science 323:116–119

Denis V, Debreuil J, De Palmas S, Richard J, Guillaume MM, Bruggemann JH (2011) Lesion regeneration capacities in populations of the massive coral *Porites lutea* at Réunion Island: environmental correlates. Mar Ecol Prog Ser 428:105–111

Frydl P (1979) The effect of parrotfish (Scaridae) on coral in Barbados, W.I. Int Rev Gesamten Hydrobiol 64:737–748

Goatley CHR, Bellwood DR (2011) The role of dimensionality, canopies and complexity in ecosystem monitoring. PLoS One 6:e27307

Harrell Jr FE (2014) Package "rms". Department of Biostatistics, Vanderbilt University, USA. http://biostat.mc.vanderbilt.edu/wiki/Main/RmS

Harrison PL, Wallace CC (1990) Reproduction, dispersal and recruitment of scleractinian corals. In: Dubinsky Z (ed) Ecosystems of the world: Coral reefs. Elsevier Science Publishers, Amsterdam, pp 133–207

Henry L, Hart M (2005) Regeneration from injury and resource allocation in sponges and corals - a review. Int Rev Gesamten Hydrobiol 90:125–158

Hughes TP, Rodrigues MJ, Bellwood DR, Ceccarelli D, Hoegh-Guldberg O, McCook L, Moltschaniwskyj N, Pratchett MS, Steneck RS, Willis B (2007) Phase shifts, herbivory, and the resilience of coral reefs to climate change. Curr Biol 17:360–365

Jayewardene D, Donahue M, Birkeland C (2009) Effects of frequent fish predation on corals in Hawaii. Coral Reefs 28:499–506

Littler MM, Taylor PR, Littler DS (1989) Complex interactions in the control of coral zonation on a Caribbean reef flat. Oecologia 80:331–340

Meesters EH, Noordeloos M, Bak RPM (1994) Damage and regeneration: links to growth in the reef building coral *Montastrea annularis*. Mar Ecol Prog Ser 112:119–128

Mumby PJ (2009) Phase shifts and the stability of macroalgal communities on Caribbean coral reefs. Coral Reefs 28:761–773

Oren U, Benayahu Y, Loya Y (1997) Effect of lesion size and shape on regeneration of the Red Sea coral *Favia favus*. Mar Ecol Prog Ser 146:101–107

Rotjan RD, Lewis SL (2005) Selective predation by parrotfishes on the reef coral *Porites astreoides*. Mar Ecol Prog Ser 30:193–201

Rotjan RD, Lewis SM (2008) Impact of coral predators on tropical reefs. Mar Ecol Prog Ser 367:73–91

Sánchez JA, Gil MF, Chasqui LH, Alvarado EM (2004) Grazing dynamics on a Caribbean reef-building coral. Coral Reefs 23:578–583

Sano M, Shimizu M, Nose Y (1984) Food habits of teleostean reef fishes in Okinawa Islands, southern Japan. The Univ Mus The Univ Tokyo Bull 25:1–128

Szmant-Froelich AM (1985) The effect of colony size on the reproductive ability of the Caribbean coral *Montastrea annularis*. Proc 5th Int Coral Reef Symp 4:295–300

Shantz A, Stier A, Idjadi J (2011) Coral density and predation affect growth of a reef-building coral. Coral Reefs 30:363–367

Szmant AM, Gassman NJ (1990) The effects of prolonged "bleaching" on the tissue biomass and reproduction of the reef coral *Montastrea annularis*. Coral Reefs 8:217–224

Team RDC (2008) R: A language and environment for statistical computing. 3.0.1 ed. R Foundation for Statistical Computing, Viena

Trapon ML, Pratchett MS, Hoey AS (2013) Spatial variation in abundance, size and orientation of juvenile corals related to the biomass of parrotfishes on the Great Barrier Reef. Australia. PLoS ONE 8:e57788

Van Veghel MLJ, Bak RPM (1994) Reproductive characteristics of the polymorphic Caribbean reef building coral *Montastrea annularis* III. Reproduction in damaged and regenerating colonies. Mar Ecol Prog Ser 109:229–233

van Woesik R (1998) Lesion healing on massive *Porites* spp. corals. Mar Ecol Prog Ser 164:213–220

Welsh JQ, Bellwood DR (2012) Spatial ecology of the steephead parrotfish (*Chlorurus microrhinos*): an evaluation using acoustic telemetry. Coral Reefs 31:55–65